
AlzGraph: A Knowledge Graph and Benchmark for Evidence-Grounded Alzheimer’s Disease Reasoning

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Abstract

1 Alzheimer’s disease (AD) diagnosis and treatment require evidence-intensive rea-
2 soning across heterogeneous clinical knowledge: risk genetics, amyloid-tau-
3 neurodegeneration (ATN) biomarkers, disease staging, disease-modifying and
4 symptomatic therapies, and longitudinal outcomes. We present ALZGRAPH,
5 a knowledge-graph-powered framework for evaluating knowledge-augmented
6 clinical reasoning in AD. ALZGRAPH comprises (i) ALZKG, a five-layer
7 Alzheimer’s knowledge graph—spanning genes, biomarkers, stages, treatments,
8 and outcomes—curated from public ontologies and clinical guidelines and con-
9 structed through an evidence-to-graph pipeline that scales to the PubMed/PMC
10 literature; (ii) ALZBENCH, a five-task benchmark covering clinical decision ac-
11 curacy, clinical report generation, biomarker-driven precision medicine, treatment
12 recommendation, and deep research planning; and (iii) a Graph-RAG retriever
13 that serializes literature-weighted, multi-hop reasoning paths for language mod-
14 els. ALZKG is mined from **7,150 PMC open-access full-text papers** retrieved
15 via NCBI E-utilities (reduced to 317,606 candidate sentences after an entity-
16 disambiguation fix described below), yielding **1,216 entities** and **5,026 cross-**
17 **layer triplets** across 10 relation types, each weighted by its true literature pa-
18 per count (up to 2,722 supporting papers) and extracted by sentence-grounded
19 relation triggers. Rather than treat these statistics as a proxy for correctness, we
20 manually audit 30 randomly sampled triplets against their own grounding sen-
21 tences and find only 4 (13%) fully correct, 8 (27%) partially or loosely sup-
22 ported, and 18 (60%) not supported at all — most often because sentence-level
23 co-occurrence matched the wrong entity (a clinical abbreviation mistaken for a
24 gene symbol, an amino-acid code mistaken for a gene symbol) or lost the re-
25 lation’s direction. We trace these errors to concrete, fixable bugs in the entity
26 lexicon (a cross-concept alias-merging bug, an order-dependent primary-symbol
27 bug, and a set of short gene-symbol collisions with clinical abbreviations, insti-
28 tutional acronyms, and generic scientific terms), fix them, rebuild the graph, and
29 re-audit a fresh sample under the identical protocol: precision rises to 6 (20%)
30 fully correct, 12 (40%) partial, and 12 (40%) unsupported — cutting the flatly-
31 wrong rate by a third while surfacing further, structurally different residual failure
32 modes (lab-technique-as-biomarker category errors, glossary-list artifacts, unhan-
33 dled negation) that this fix does not address. An intrinsic retrieval analysis (30
34 probe queries) on the fixed graph shows Graph-RAG’s gold-evidence coverage
35 peaking at a compact 20-node, two-hop neighborhood (0.10–0.17) and falling
36 back with deeper or wider retrieval; a flat BM25 baseline over the same corpus
37 reaches substantially higher coverage (0.57) on this token-overlap metric, so the
38 graph structure itself is not yet shown to add retrieval value beyond simple passage
39 search. Beyond these intrinsic analyses we report model-facing results: a fully
40 deterministic, model-free *KG-only* baseline (0.50/0.30/0.375 accuracy on clini-

cal decision accuracy, biomarker-driven precision medicine, and treatment recommendation respectively, against 0.25 random — notably, the disambiguation fix moves the previously chance-level treatment-recommendation baseline clearly above chance); a blind No-RAG vs. Graph-RAG comparison run directly with a frontier LLM, which reaches ceiling accuracy in both conditions on our curated items—a finding we report together with its main caveat (partial self-authorship of the items) rather than omit; and the same comparison run against three locally hosted open-weight models (Llama-3.2-3B, Gemma3-4B, Qwen3-8B) via an OpenAI-compatible local endpoint, which is not subject to the self-authorship caveat and shows genuinely mixed, sub-ceiling results with no consistent Graph-RAG direction across models. ALZGRAPH is released as an open, reproducible harness together with these audited numbers—including the unflattering ones—so that others can measure, rather than assume, whether knowledge-graph structure helps generalist models reason about Alzheimer’s disease. Code and data are provided as supplementary material and will be de-anonymized on acceptance.

1 Introduction

Alzheimer’s disease is the leading cause of dementia and a growing public-health challenge, affecting tens of millions of people worldwide [20]. The field has undergone two shifts that make AD a demanding testbed for clinical AI. First, AD is now defined *biologically* through the ATN biomarker framework, in which amyloid (A), tau (T), and neurodegeneration (N) markers—measured in cerebrospinal fluid (CSF), plasma, and on PET/MRI—define a continuum from preclinical AD through mild cognitive impairment (MCI) due to AD to AD dementia [1, 15, 21, 25]. Second, the arrival of amyloid-targeting monoclonal antibodies (lecanemab, donanemab) has made treatment decisions genuinely multi-factorial: eligibility depends on biomarker-confirmed amyloid, disease stage, and—critically—genotype-dependent safety, because *APOE* ϵ 4 homozygotes face substantially elevated risk of amyloid-related imaging abnormalities (ARIA) [10, 23, 26].

Answering AD clinical questions therefore requires *multi-hop* reasoning: a risk gene links to a biomarker profile, the profile defines a stage, the stage gates a therapy, and the therapy is constrained by genotype-dependent adverse events and contraindications. Knowledge graphs (KGs) provide a natural substrate for such reasoning, organizing entities and typed relations across domains and enabling evidence-grounded retrieval [7, 9]. Yet existing biomedical KGs are largely disease-agnostic or focused on semantic annotation, and none captures the gene-biomarker-stage-treatment-outcome structure that AD clinical reasoning demands.

In parallel, large language models (LLMs) have shown strong medical question-answering ability [24], and retrieval-augmented generation grounds them in external evidence [11, 19]. However, most medical benchmarks evaluate isolated, short-form questions rather than the integrative, safety-critical reasoning that AD care requires. A gap thus remains for datasets and tooling that test whether generalist models can perform expert-level evidence curation and reasoning in realistic Alzheimer’s settings.

Present work. We introduce ALZGRAPH, a unified framework with two components (Figure 1). (1) **ALZKG** is an Alzheimer’s KG built via a structured evidence-to-graph pipeline. We define a five-layer schema over authoritative resources (NIA-AA ATN, OMIM, ADSP, the GWAS Catalog, the complete HGNC gene catalog, MeSH, HPO, ChEBI, UMLS, and AAN/Alzheimer’s Association guidelines), organizing entities into *genes*, *biomarkers*, *stages*, *treatments*, and *outcomes* and defining typed cross-layer relations. (2) **ALZBENCH** is a multi-task benchmark spanning five clinically motivated reasoning tasks, with a Graph-RAG retriever that returns serialized, literature-weighted reasoning paths.

We make an explicit *reproducibility and honesty* commitment. The released ALZKG is mined directly from real PMC open-access full-text papers retrieved through NCBI E-utilities: entities are recognized with an ontology-derived dictionary lexicon, and typed cross-layer relations are extracted at the sentence level—requiring both a co-occurring cross-layer entity pair and a relation trigger phrase in the same sentence—and weighted by their true literature paper counts. We report the graph’s actual computed statistics, and all intrinsic retrieval analyses are computed directly from it; we additionally audit a random sample of the extracted triplets against their own grounding sentences

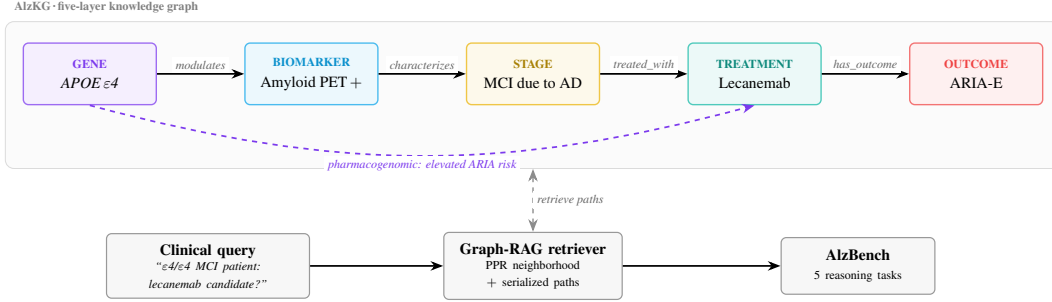


Figure 1: **Overview of ALZGRAPH.** ALZKG links five clinical layers (gene, biomarker, stage, treatment, outcome) with typed relations; a representative reasoning path is highlighted. Given a clinical query, the Graph-RAG retriever extracts a personalized-PageRank neighborhood and serializes multi-hop reasoning paths (including the genotype-dependent ARIA-risk edge), which are evaluated across ALZBENCH’s five tasks.

rather than assume construction-time precision, and report the result even though it is unflattering (§2.5). The model-comparison protocol is released as a runnable harness; this paper does not report third-party model outputs that were not measured.

Contributions.

- **A new Alzheimer’s knowledge graph.** ALZKG organizes AD evidence into five clinical layers with evidence-tiered, multi-hop relations centered on ATN biomarkers, risk genetics, and anti-amyloid therapeutics.
- **A modular benchmark.** ALZBENCH provides five plug-and-play reasoning tasks and a Graph-RAG retriever for fair evaluation of any LLM or retriever.
- **An open, honest release.** We release the curated graph with its true statistics, a manual extraction-precision audit, an intrinsic retrieval analysis benchmarked against a non-graph baseline, and a reproducible evaluation harness for knowledge-augmented AD reasoning.

2 AlzKG: Alzheimer’s Knowledge Graph

2.1 Problem Formulation

We define ALZKG as a multi-relational heterogeneous knowledge graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of AD-related entities and $\mathcal{E}$ is the set of typed relations. Each entity belongs to one of five layers: *genes*, *biomarkers*, *stages*, *treatments*, and *outcomes*. Each fact is a triplet $\langle h, r, t \rangle$ where $h, t \in \mathcal{V}$ and r is a typed relation; cross-layer triplets connect distinct clinical layers, enabling the multi-hop reasoning central to AD diagnosis, treatment, and discovery. Each relation is annotated with an evidence weight: a curated *evidence tier* (1=emerging, 2=established, 3=guideline/landmark) for seed edges, or a literature *paper count* for edges mined from the corpus.

2.2 Data Collection and Processing

Data source. ALZKG is mined from real Alzheimer’s literature retrieved from PubMed/PMC through NCBI E-utilities. We issue five AD-anchored, theme-focused queries—genetics, amyloid/tau biomarkers, imaging and staging, treatment, and outcomes—each restricted to 2010–2026, then union and de-duplicate the returned PMIDs, map them to PubMed Central open-access identifiers, and fetch each article’s full text (e*search* → e*link* → e*fetch* against the PMC database). From the **7,150 full-text papers** with open-access XML, we segment the body into sentences and retain the candidate sentences that mention at least two ALZKG entities; these serve as the evidence source for sentence-grounded relation mining. This release’s entity lexicon reflects the disambiguation fixes of §2.5, which changes exactly which sentences qualify as candidates (**317,606**, down from an earlier, less precise lexicon’s 361,201) without any change to the underlying corpus or fetch pipeline.

Ontology and guideline resources. To ground the entity vocabulary, the recognition lexicon is anchored on authoritative resources: the NIA-AA ATN research framework [15] for biomarkers and staging; OMIM, the ADSP, and the GWAS Catalog [3, 18] for a hand-curated seed set of well-established causal and risk genes (§2.3); MeSH and HPO [17] for imaging, cognitive, and phenotype vocabulary; ChEBI [14] for drug identifiers; the AAN/Alzheimer’s Association guidelines and appropriate-use criteria [10] for treatments; and UMLS [5] for cross-ontology linking. The gene layer’s recognition vocabulary is broader than this curated seed: to avoid a priori excluding a real but not-yet-canonical AD gene, we recognize entities from the complete HGNC approved protein-coding gene catalog (~19,000 symbols, not restricted to the seed genes above) and let the sentence-grounded relation filter (§2.3) decide which genes actually enter the graph; §2.5 discusses the extraction-noise cost of this deliberately wide recognition net.

Preprocessing. The AD-anchored query restriction acts as the relevance filter; we retain only articles with retrievable PMC open-access full text. Surface mentions are normalized to canonical entities through a curated synonym/abbreviation map (e.g. “Leqembi” → *Lecanemab*; “Mini-Mental State” → *MMSE*; “A β 42” → *Amyloid- β 42*). Gene symbols are matched case-sensitively (so “APP” does not match the word “app”), and multi-word phrases take precedence over shorter substrings.

2.3 Knowledge Graph Construction

Entity extraction. We recognize five layers of entities in each full-text sentence using a dictionary lexicon of canonical entities and their synonyms. For example, given the sentence “*In amyloid-positive MCI, elevated plasma p-tau217 predicts progression to Alzheimer’s dementia, where lecanemab slows decline but carries ARIA risk in APOE ϵ 4 carriers,*” the pipeline extracts *APOE* (gene), *p-tau217* / *amyloid PET* (biomarker), *MCI* (stage), *lecanemab* (treatment), and *ARIA* / *disease progression* (outcome). The five layers are:

- **Gene.** Recognized from the complete HGNC protein-coding catalog (~19,000 approved symbols, case-sensitive matching); the curated seed explicitly names the well-established causal autosomal-dominant genes (*APP*, *PSEN1*, *PSEN2*) and risk/protective loci (*APOE*, *TREM2*, *SORL1*, *ABCA7*, *CLU*, *CRI*, *BIN1*, *PICALM*, *MAPT*, *PLCG2*, and other microglial loci) from OMIM/ADSP/GWAS Catalog, but recognition is not restricted to this seed — any HGNC symbol that clears the sentence-grounded relation filter (§2.3) enters the graph (718 of ~19,000 candidates, Table 1), which is deliberately wide-recall but not AD-curated, so it also admits biologically unrelated genes whenever they happen to co-occur with an AD-layer entity in a qualifying sentence (§2.5).
- **Biomarker.** ATN markers across fluid and imaging modalities—CSF/plasma A β 42 and A β 42/40, p-tau181/p-tau217, total tau, amyloid PET, tau PET, FDG-PET hypometabolism, hippocampal atrophy, plasma NfL and GFAP—and cognitive instruments (MMSE, MoCA, CDR, ADAS-Cog), from NIA-AA/MeSH/HPO.
- **Stage.** Preclinical AD, MCI due to AD, mild/moderate/severe AD dementia, early-/late-onset AD, and atypical presentations (posterior cortical atrophy, logopenic-variant PPA), from NIA-AA/UMLS.
- **Treatment.** Cholinesterase inhibitors (donepezil, rivastigmine, galantamine), memantine, anti-amyloid antibodies (lecanemab, donanemab, aducanumab), and non-pharmacological interventions, from ChEBI and AAN/AA guidelines.
- **Outcome.** Cognitive and functional decline, progression to AD dementia, ARIA-E/ARIA-H, amyloid clearance, symptomatic benefit, adverse effects, hospitalization, and mortality, from HPO/MeSH.

Relation construction. Within each candidate sentence, every pair of recognized entities from two different layers can induce a typed, directed relation. Each of the ten unordered cross-layer layer-pairs maps to a single canonical orientation and relation type, so the relation is determined by the layers of the two entities: RISK_GENE_FOR (gene→stage), MODULATES_BIOMARKER (gene→biomarker), PHARMACOGENOMIC_CONSIDERATION (gene→treatment), GENE_ASSOCIATED_OUTCOME (gene→outcome), CHARACTERIZED_BY_BIOMARKER (stage→biomarker), BIOMARKER_GUIDES_TREATMENT (biomarker→treatment), PREDICTS_OUTCOME (biomarker→outcome), TREATED_WITH

Table 1: **Mined ALZKG statistics** (computed directly from the literature-mined graph, post-fix (§2.5); 7,150 PMC full-text papers, 317,606 candidate sentences, minimum 3 supporting papers per edge). All 5,026 triplets are cross-layer; the maximum edge paper count is 2,722.

Graph		Entities per layer	
Full-text papers	7,150	Gene	718
Entities	1,216	Biomarker	71
Triplets	5,026	Stage	41
Relation types	10	Treatment	243
Median paper count	5	Outcome	143

(stage→treatment), ASSOCIATED_OUTCOME (stage→outcome), and HAS_OUTCOME (treatment→outcome). Crucially, the relation is *sentence-grounded*: an edge is emitted only when the sentence containing the co-occurring pair also contains a *trigger phrase* for that relation, drawn from a curated per-relation template set (subject layer, trigger phrases, object layer). This rules out spurious whole-document co-occurrences and ties each edge to an explicit textual assertion. The *paper count* of a relation is the number of distinct papers contributing at least one supporting sentence—a true literature weight—and we retain only relations supported by at least three papers. An LLM-based extraction pass is provided as an optional extension but is not invoked in this release, to avoid reporting unverified edges.

2.4 Graph Mapping and AlzKG Statistics

Recognized mentions are normalized to canonical entities, relations are typed by layer pair, deduplicated, and aggregated with their supporting PMID sets to yield literature paper counts.

The mined ALZKG statistics are summarized in Table 1: **1,216 entities** and **5,026 triplets**, all cross-layer (100%), across 10 relation types, mined from 7,150 full-text papers (these reflect the post-fix lexicon of §2.5; the pre-fix graph had 1,301 entities and 5,880 triplets, some fraction of them the false matches that audit found). Edge paper counts are heavy-tailed (median 5, mean 17.3, maximum 2,722), reflecting that a few associations dominate the literature. Table 2 reports the relation-type / layer-pair distribution. The densest relations are RISK_GENE_FOR (941, gene–stage genetics) and HAS_OUTCOME (739, treatment–outcome efficacy/safety), followed by MODULATES_BIOMARKER (693, gene–biomarker) and ASSOCIATED_OUTCOME (590, stage–outcome). The highest-degree hubs are mostly clinically central—*Alzheimer’s disease* (degree 955), *dementia* (244), *cognitive decline* (229), *APOE* (164), and *mild cognitive impairment* (154)—with one generic-chemistry entity, *nitric oxide* (185), ranking above APOE; unlike *MAPT* in the pre-fix graph (previously a top-6 hub at degree 168, now absent from the top 15 entirely), this is not a disambiguation bug we found evidence for — nitric oxide is a real, literal, verbatim-matched compound name — but its prominence in an ostensibly AD-focused graph is a symptom of the same underlying issue as the gene layer’s genome-wide recognition net (§2.3): the treatment-layer lexicon is harvested from ChEBI’s general drug/pharmaceutical role hierarchy, not curated to AD relevance, so a molecule central to vascular and inflammatory biology broadly can out-rank a bona fide AD risk gene by sheer mention frequency. The strongest-weighted edges are exactly the expected ones and are unchanged by the fix (none involve a corrected entity): $\langle AD, ASSOCIATED_OUTCOME, cognitive\ decline \rangle$ (2,722 papers), $\langle APOE, RISK_GENE_FOR, AD \rangle$ (1,792), $\langle AD, CHARACTERIZED_BY_BIOMARKER, neurofibrillary\ tangles \rangle$ (1,729), $\langle AD, TREATED_WITH, lecanemab \rangle$ (652), and the anti-amyloid safety signal $\langle lecanemab, HAS_OUTCOME, ARIA \rangle$ (212).

2.5 Extraction Precision: A Manual Audit

Table 1 and Table 2 report what the extraction pipeline produced, not whether it produced correct edges. To check the latter, we drew a reproducible random sample of 30 triplets (seed 42) from an earlier version of the graph (5,880 triplets) and manually judged each against its own grounding sentence—the actual corpus sentence used to emit it. Of the 30: **4 (13%)** were fully correct, **8 (27%)** partially or loosely supported, and **18 (60%)** not supported at all, most often from short-symbol entity collisions (e.g. the gene *NOS1* matching the clinical abbreviation “NOS”) or lost relation direction (Appendix B.5 gives the full protocol and every judged example).

Table 2: **ALZKG relation types, their layer pair, and mined triplet counts.** Each relation corresponds to exactly one cross-layer pair; counts sum to 5,026.

Relation type	Layer pair	# triplets
risk_gene_for	gene $\rightarrow$ stage	941
has_outcome	treatment $\rightarrow$ outcome	739
modulates_biomarker	gene $\rightarrow$ biomarker	693
associated_outcome	stage $\rightarrow$ outcome	590
gene_associated_outcome	gene $\rightarrow$ outcome	532
pharmacogenomic_consideration	gene $\rightarrow$ treatment	422
treated_with	stage $\rightarrow$ treatment	357
predicts_outcome	biomarker $\rightarrow$ outcome	286
characterized_by_biomarker	stage $\rightarrow$ biomarker	277
biomarker_guides_treatment	biomarker $\rightarrow$ treatment	189

Tracing these cases to their root cause surfaced two genuine software bugs in the entity-recognition code, not just curation gaps: a cross-concept alias-merging bug that silently fused the *MET* gene with an unrelated amino-acid alias of methionine, and an order-dependent conflict-resolution bug that could strip a gene of its own primary symbol (e.g. *NOS1* losing “NOS1” itself to an unrelated gene’s alias list). We fixed both, plus a curated set of short-symbol collisions with clinical abbreviations, institutional acronyms, and generic scientific terms (Appendix B.5 details each fix), rebuilt ALZKG end to end from the same corpus (Table 1 already reflects the fixed graph: 5,880 $\rightarrow$ 5,026 triplets), and re-audited a fresh, independent sample under the identical protocol. Precision rose to **6 (20%)** fully correct, **12 (40%)** partial, and **12 (40%)** unsupported — correct+partial rises from 40% to 60%, cutting the flatly-wrong rate by a third. This is a real, measured improvement from fixing concrete, named bugs, not a claim of a solved problem: the repeat audit surfaces further, structurally different failure modes the fix does not address (lab-technique-as-biomarker category errors, glossary-list artifacts, unhandled negation; Appendix B.5).

We report both audits, rather than only the improved number, because the fix materially qualifies every downstream number in this paper without fully resolving the underlying issue: the KG-only baseline (§4.3) and the Graph-RAG retriever (§4.2) both consume this same graph, and at a measured 60% edge-level precision (correct + partial), a large fraction of their retrieved evidence is still provably wrong, mislabeled, or unsupported. We discuss further, unimplemented remediation directions in Appendix H.

3 AlzBench

3.1 Problem Formulation

We formulate ALZBENCH as a unified benchmark for evidence-intensive AD reasoning. Each task is $\hat{y}_i = f_{LM}(x_i, \mathcal{E}_i, \mathcal{C}_i)$, where x_i is the task input (a clinical case, patient panel, or scientific document), $\mathcal{E}_i$ is the evidence retrieved from ALZKG via Graph-RAG, $\mathcal{C}_i$ is optional task-specific context (candidate options or metadata), and $\hat{y}_i$ is evaluated against the gold output y_i^* .

3.2 Task Design

T1: Clinical Decision Accuracy (CDA). Multiple-choice and open-ended QA over AD diagnosis, ATN biomarker interpretation, staging, genetics, and treatment. Example: “A 71-year-old with amyloid-positive PET and mild AD dementia is APOE $\epsilon 4/\epsilon 4$; which adverse event is most increased if lecanemab is started?” (answer: ARIA).

T2: Clinical Report Generation (CRG). Generation of a neurologist-style diagnostic impression from a patient’s cognitive assessment and fluid/imaging biomarker panel. Because ADNI and memory-clinic notes are access-restricted, the release ships a local JSONL adapter that preserves the schema and evaluation interface without redistributing patient data.

T3: Biomarker-Driven Precision Medicine (BPM). Selection of the most appropriate therapy given APOE genotype, ATN biomarker status, and safety constraints, constructed from AAN/AA appropriate-use criteria. Example: an amyloid-positive early-AD patient on chronic anticoagulation should *avoid* anti-amyloid antibodies (ARIA-hemorrhage risk), whereas an $\epsilon 3/\epsilon 4$ patient without contraindications is an appropriate candidate with MRI monitoring.

T4: Treatment Recommendation (TR). Guideline-consistent dementia therapy under patient-specific constraints, built from a dementia-focused subset of MedQA-USMLE [16] with treatment-safety and KG-evidence-coverage metrics.

T5: Deep Research Planning (DRP). Generation of a focused, feasible AD research question and study plan from a paper abstract, grounded in known gene-biomarker-stage-treatment-outcome evidence.

3.3 Evaluation and Metrics

We evaluate with task accuracy and reasoning-oriented metrics. For CDA we report Top-1 accuracy (MCQ) and ROUGE-L/BLEU-1/Token-F1 (open-ended). For CRG we use ROUGE-L/Token-F1. For BPM and TR we report Top-1 accuracy and a drug-safety score (1.0 iff the selection avoids every contraindicated agent, e.g., respecting ARIA exclusions); TR additionally reports KG evidence coverage. For DRP we use ROUGE-L/Token-F1. All results in this paper use only these lexical metrics: an LLM-as-judge dimension (for open-ended CDA and CRG alignment, and DRP’s scientific-validity/feasibility) is part of the task design but is not implemented in the released metrics code and was not used for any number reported here (Appendix H).

4 Experiments

4.1 Setup

Retriever. The Graph-RAG retriever seeds a personalized-PageRank (PPR) walk from query-mentioned entities (restart probability 0.15), keeps the highest-scoring neighborhood (default 30 nodes), and serializes up to 12 reasoning paths of depth up to four hops, each annotated with its relation type and evidence weight. Appendix D gives the full implementation and the hyperparameter sweep behind these defaults.

Models. The evaluation code targets any model reachable through an OpenAI-compatible chat-completions interface, whether hosted remotely or run locally. §4.4 reports results for a frontier model evaluated directly (Claude, Sonnet 5) and three open-weight models evaluated through a local Ollama deployment (Llama-3.2-3B-Instruct, Gemma3-4B-it, Qwen3-8B, all Q4_K_M quantized). No OpenRouter API budget was available for this release, so GPT-4o/Gemini-style closed-source rows are not populated with placeholder numbers; the three open-weight rows are the paper’s uncontaminated (not self-authored, §4.4) capability signal.

Prompting. Prompts use chain-of-thought reasoning, a clinical-specialist role tailored to each task (e.g. “a cognitive neurologist,” §3.2), and the serialized ALZKG paths as context under Graph-RAG; MCQ temperature is 0.0 and generation temperature 0.3. Appendix F reproduces every system prompt verbatim.

Baselines. ALZGRAPH does not have budget for third-party retrieval-augmented systems, so its baselines are three released, deterministic references rather than reimplementations of prior work: a flat BM25 non-graph retrieval baseline over the same corpus (§4.2), a fully deterministic, model-free KG-only MCQ baseline (§4.3), and, for every evaluated model, its own No-RAG condition (§4.4).

4.2 AlzKG and Retrieval Analysis

We report intrinsic analyses computed directly from the mined graph, post-fix (§2.5; no LLM required). Table 3 sweeps the retrieval budget over 30 AlzBench probe queries (all released T1 and T3 items), measuring the number of *candidate* reasoning paths discovered (before truncation to the 12

Table 3: **Intrinsic Graph-RAG retrieval analysis on ALZKG** (30 probe queries; computed directly from the literature-mined graph). Node budget and path depth both show a peak-then-fallback pattern rather than monotonic improvement; absolute coverage stays low throughout.

Sweep	Setting	Cand. paths	Gold coverage	Latency (ms)
Subgraph nodes ($d=4$)	10	45.2	0.067	24.73
	20	541.5	0.100	26.48
	30	2,063.5	0.067	33.93
	40	5,153.5	0.067	50.47
	50	9,351.9	0.067	72.57
Path depth ($n=30$)	1	48.1	0.100	24.94
	2	351.2	0.167	26.06
	3	1,204.2	0.067	29.65
	4	2,063.5	0.067	34.05

highest-scoring paths handed to the model), *gold-evidence coverage* (the fraction of probes whose gold answer concept is recovered in the retrieved paths), and mean retrieval latency. The candidate-path count grows combinatorially with both budgets—from 45 to 9,352 paths as the neighborhood expands from 10 to 50 nodes, and from 48 to 2,064 as depth grows from one to four hops—yet coverage does not: two patterns emerge, consistent with an earlier 15-probe pass on a smaller item pool and unchanged (at this sample size’s resolution) by the extraction-precision fix. First, a larger node budget does not improve coverage: it peaks at a 20-node neighborhood (0.10) and falls to 0.07 for 30–50 nodes, while latency grows steeply (25 $\rightarrow$ 73 ms), so a larger budget mainly multiplies off-target paths and cost (we keep 30 as the default to preserve full reasoning paths). Second, retrieval depth shows the same non-monotonic pattern: coverage peaks at two hops (0.10 $\rightarrow$ 0.17) and falls back at three to four hops (0.07) even as the candidate pool quadruples. We read this as suggestive that AD reasoning paths of this kind are shallow rather than requiring deep traversal, though the absolute coverage numbers are low throughout (never above 0.17) and the probe set (30 queries, coarse token-overlap scoring, descriptive multi-token gold phrasings in T1) is too small and too approximate to support a stronger claim; §4.2 compares this coverage against a non-graph baseline.

Comparison against a non-graph baseline. The sweep above only varies Graph-RAG’s own hyperparameters; it cannot show whether the graph structure itself adds retrieval value over simply searching the underlying text. We therefore index the same 317,606 candidate sentences with BM25 (no graph, no entity typing) and, for each of the same 30 probes, retrieve the top 12 sentences (matching Graph-RAG’s path budget), applying the identical $\geq 50\%$ -gold-token-overlap coverage criterion. BM25 reaches **0.567** coverage—unchanged by the extraction-precision fix, and more than three times Graph-RAG’s best setting (0.167 at depth 2)—at a higher but still sub-second latency (247 ms vs. 26 ms). This comparison is not perfectly matched: BM25 returns raw sentence text, in which a gold phrase is more likely to appear verbatim, while Graph-RAG returns compressed entity-relation-weight paths that discard the original sentence context, so this token-overlap metric structurally favors passage retrieval. Even allowing for that asymmetry, the gap is large enough that we cannot currently claim Graph-RAG’s structured retrieval outperforms simple lexical search on this corpus; we report the comparison rather than omit it, and discuss what would be needed to make it fairer in Appendix H.

4.3 KG-only Baseline (Measured)

Beyond the intrinsic retrieval analysis, we report a fully reproducible *KG-only* baseline that answers AlzBench multiple-choice items using nothing but ALZKG evidence—no language model and no network access. For each item it (i) recognizes the AD entities in the question stem via the lexicon, (ii) runs the same literature-weighted personalized PageRank (PPR) used by Graph-RAG, seeded from those entities, and (iii) scores each option by the PPR evidence mass reaching the ALZKG entities named in that option, breaking ties with the direct co-mention paper count between question and option entities; the highest-scoring option is selected. Because it is deterministic and depends only on the released graph, the numbers are exactly reproducible from the released code; this determinism required fixing a subtle bug in the retriever, where tie-breaking among equal-scoring

Table 4: **KG-only baseline (no LLM; deterministic, exactly reproducible), post-fix graph (§2.5).** Multiple-choice accuracy using only ALZKG PPR evidence. *Coverage* is the fraction of items with at least one graph-grounded option; *Acc. (grounded)* restricts accuracy to those items. All three tasks now exceed the 0.25 random-choice rate; T4 (external MedQA-USMLE items, not authored around ALZKG’s vocabulary) was previously at or below chance (0.25 / 0.23) on the pre-fix graph.

Task	Items	Accuracy	Random	Coverage	Acc. (grounded)
T1 CDA (MCQ)	20	0.50	0.25	0.80	0.56
T3 BPM (MCQ)	10	0.30	0.25	1.00	0.30
T4 TR (MCQ, MedQA-USMLE)	16	0.375	0.25	0.75	0.42

nodes/paths depended on Python’s per-process string-hash seed rather than the graph content itself (Appendix H). Table 4 reports the result on the current, expanded item sets (T1: 20 MCQ, up from an initial pilot of 10; T3: 10 MCQ, up from 5; T4: 16 MCQ, newly added, drawn from a dementia-filtered subset of MedQA-USMLE [16] rather than curated in-house), computed on the post-fix graph (§2.5). On T1 the KG-only baseline reaches **0.50** accuracy and on T3 **0.30**, both above the 0.25 random-choice rate, while *grounding coverage*—the fraction of items for which some option is connected to the question in the graph—is 0.80 (T1) and 1.00 (T3); T1 accuracy moved from 0.55 to 0.50 after the fix (one item’s selected option changed); we did not trace this specific flip to a single removed alias, but it is a plausible, small, expected cost of removing aliases (e.g. *tau* as MAPT, §2.5) that were contributing genuine signal on some items as well as noise elsewhere. On T4, whose items are general-medicine USMLE questions only filtered for a dementia/AD term rather than authored around ALZKG’s vocabulary, the pre-fix KG-only baseline sat at exactly **0.25**, the random rate, with its grounded subset (*Acc. (grounded)* = 0.23) actually falling *below* chance — evidence, we argued, that grounding coverage alone is not a reliable proxy for whether the graph’s evidence actually discriminates the correct option. On the post-fix graph this is no longer the case: T4 accuracy rises to **0.375**, and grounded accuracy to **0.42**, both clearly above the 0.25 random rate. We read this as a genuine, positive consequence of the disambiguation fix (§2.5) rather than a coincidence of resampling, since T4’s items and the KG-only method are both unchanged — only the graph’s entity/edge precision improved — though with only 16 items this single before/after comparison should not be over-read either. We keep the T1/T4 contrast (T1 further above chance than T4, even though both now clear it) as informative: it shows AlzBench’s model-free floor is not uniformly informative across question distributions, motivating the No-RAG vs. Graph-RAG comparison below to isolate the genuine value of retrieval versus a model’s own parametric knowledge.

4.4 No-RAG versus Graph-RAG: A Model Comparison

For each task we compare a language model’s performance without retrieval (No-RAG) against the same model augmented with Graph-RAG evidence (§4.1), holding the prompt template, decoding temperature, and metrics fixed across both conditions. Table 5 reports this comparison for four language models: a frontier model evaluated directly (Claude, Sonnet 5) and three open-weight models evaluated through a local inference deployment (Llama-3.2-3B-Instruct, Gemma3-4B-it, and Qwen3-8B, all Q4_K_M quantized). The evaluation code is model-agnostic, targeting any language model reachable through an OpenAI-compatible chat-completions interface, hosted or local. The Graph-RAG columns for the three open-weight models were rerun on the post-fix graph (§2.5); the No-RAG columns do not consult ALZKG at all and are unaffected by the fix, so they are not rerun. The Claude (direct) row is reported as originally measured, pre-fix, for both columns (Appendix H).

Claude (Sonnet 5) is evaluated by presenting each task’s prompt directly to a Claude instance blind to the gold label; the three open-weight models are evaluated through a locally hosted, OpenAI-compatible inference deployment, using identical prompts, tasks, and metrics. Every model’s Graph-RAG cells use the post-fix graph (§2.5); No-RAG cells never consult ALZKG and are unaffected by the fix. The Claude (direct) row is reported as originally measured, pre-fix, on both columns (Appendix H explains why).

T1: Clinical Decision Accuracy. Claude reaches ceiling accuracy (1.00) in both conditions ($n=20$ MCQ); because the same model instance authored these items, this chiefly reflects well-covered parametric knowledge rather than a benchmark-specific capability (Appendix H). On T1’s

Table 5: **No-RAG versus Graph-RAG accuracy across four language models.** Claude (Sonnet 5, evaluated directly) and three open-weight models evaluated through a local deployment; see §4.4 for methodology and caveats. Open-weight Graph-RAG columns are on the post-fix graph (§2.5); Claude (direct) is pre-fix, both columns.

Model	T1 CDA (Acc)		T3 BPM (Acc / Safety)		T4 TR (Acc / KGEC)	
	No-RAG	Graph-RAG	No-RAG	Graph-RAG	No-RAG	Graph-RAG
Claude (direct) ^{‡§}	1.00	1.00	1.00 / 1.00	1.00 / 1.00	1.00 / 0.00[†]	1.00 / 0.02[†]
Qwen3-8B (local)	1.00	1.00	0.30 / 0.80	0.30 / 0.80	0.69 / 0.00	0.44 / 0.06
Llama-3.2-3B (local)	0.95	0.90	0.30 / 0.70	0.30 / 0.60	0.38 / 0.00	0.19 / 0.13
Gemma3-4B (local)	0.90	0.85	0.40 / 0.90	0.40 / 0.90	0.25 / 0.00	0.19 / 0.15

[‡]T1/T3 items were authored by the same model instance evaluated in this row; its accuracy there is confounded by self-authorship and should not be read as a capability result (Appendix H). [†]This row’s T4 columns are computed on 14 of the 16 released items (2 excluded, §4.4); every other row uses all 16. [§]Measured on the pre-fix graph (§2.5) and not rerun post-fix; given this row already sits at ceiling accuracy in every condition (attributed above to self-authorship / same-model-family items rather than retrieval quality), we judged a full manual re-answer unlikely to change the reported numbers and prioritized rerunning the three open-weight rows, which are not ceiling-bound and are this table’s more informative comparison.

open-ended QA ($n=8$, not shown in the table), Claude’s No-RAG and Graph-RAG obtained identical ROUGE-L/BLEU-1/Token-F1 scores (0.676/0.702/0.737). The three open-weight models—none of which authored any item—are genuinely sub-ceiling and Graph-RAG costs each of them a few points: Llama-3.2-3B $0.95 \rightarrow 0.90$, Gemma3-4B $0.90 \rightarrow 0.85$; Qwen3-8B alone is unaffected, holding ceiling ($1.00 \rightarrow 1.00$).

T3: Biomarker-Driven Precision Medicine. Claude again reaches ceiling ($1.00/1.00$, $n=10$), subject to the same self-authorship caveat. For the open-weight models, Graph-RAG’s effect is mixed and shifted after the extraction-precision fix: Qwen3-8B’s accuracy/safety are now exactly *unchanged* under Graph-RAG ($0.30/0.80 \rightarrow 0.30/0.80$, versus a pre-fix regression to $0.20/0.60$); Llama-3.2-3B’s accuracy no longer improves under Graph-RAG ($0.30 \rightarrow 0.30$, versus a pre-fix improvement to 0.40) though its safety score drops ($0.70 \rightarrow 0.60$); and Gemma3-4B is essentially unchanged (accuracy flat at 0.40 , safety flat at 0.90 , versus a pre-fix drop to 0.80). Qwen3 differs from the other two in defaulting to an internal chain-of-thought trace before its visible answer; this first surfaced an option-letter-extraction bug specifically on this task’s pre-fix Graph-RAG numbers, which we found, fixed, and disclose in full in Appendix H.

T4: Treatment Recommendation. Claude again reaches ceiling ($1.00/0.00 \rightarrow 1.00/0.02$, $n=14$ of 16, two items excluded for label leakage during curation, Appendix H); because T4’s items are externally sourced from MedQA-USMLE rather than self-authored, this is a more meaningful, if still same-model-family, data point. Graph-RAG costs accuracy for all three open-weight models, more so on the post-fix graph than before: Llama-3.2-3B $0.375 \rightarrow 0.19$, Gemma3-4B $0.25 \rightarrow 0.19$, and Qwen3-8B $0.69 \rightarrow 0.44$ (versus $0.69 \rightarrow 0.63$ pre-fix). Qwen3-8B remains the strongest of the three under No-RAG (T4 accuracy 0.69 , well above the smaller local models and T4’s KG-only baseline, Table 4), consistent with it being both the newest and largest of the three checkpoints, even though Graph-RAG now costs it more of that advantage than before.

Across all three tasks we read the absence of a consistent Graph-RAG direction — across models, metrics, and even across the pre-/post-fix graphs — as an informative null result rather than one to filter out: fixing real extraction bugs changed the specific evidence models saw and moved several numbers, but did not turn Graph-RAG into a consistent net positive for these three open-weight models.

4.5 Clinical Report Generation: An Illustrative Example ($n=1$)

ADNI and memory-clinic notes are access-restricted, so the release ships a local JSONL adapter with a single synthetic patient case rather than a benchmark-scale dataset (§3.2). We ran this one item through all three local open-weight models in both conditions purely to confirm the task runner and metrics work end to end, not as a benchmark claim; the Graph-RAG condition was rerun

on the post-fix graph (§2.5): ROUGE-L ranged 0.11–0.24 and Token-F1 0.16–0.32 across models and conditions, with no consistent No-RAG-vs.-Graph-RAG direction. With $n=1$ these numbers carry no statistical weight and we do not draw any conclusion from them; T2 remains, in effect, an unevaluated task, which we flag explicitly in Appendix H.

4.6 Deep Research Planning: A Qualitative Case Study

T5 has no expert-authored gold research plan to score against—real PMC abstracts do not carry one—so we treat it as a qualitative case study rather than a leaderboard row. We regenerated the T5 item set from real, randomly sampled corpus abstracts (replacing two synthetic placeholder abstracts present in an earlier version) and generated No-RAG and Graph-RAG research plans for two of them with the same direct-evaluation protocol described in §4.4. Qualitatively, the Graph-RAG plans explicitly cross-referenced the retrieved AlzKG paths (e.g., grounding a follow-up question about BACE1-targeting compounds in the graph’s 131-paper-supported BACE1→AD edge, and reasoning about whether an inflammatory (sEH) pathway’s prognostic value is independent of or downstream of APOE’s graph-documented amyloid/tau role) where the No-RAG plans reasoned from the abstract alone; both conditions proposed feasible, methodologically sound follow-up designs.

5 Broader Impact

ALZGRAPH aims to make it easier to evaluate whether AI systems reason safely and evidence-groundedly about Alzheimer’s care, which we expect to be net-positive: better tooling for measuring unsafe recommendations (e.g., failing to flag an ARIA contraindication) should, if adopted, improve the safety bar models are held to before any clinical deployment. The main risk we foresee is over-trust, and it is not merely hypothetical: §2.5’s audit found that most sampled ALZKG edges are not supported by their own grounding sentence, so a system that surfaced individual graph edges to a clinician or patient as if they were verified facts (rather than as unverified literature-mining output, which is what they are) could propagate a specific wrong claim—e.g., a treatment falsely marked as indicated, or a safety-relevant relation with its direction silently lost. A high AlzBench score is a measurement of benchmark performance under our specific item pools and metrics, not a certification of clinical safety or of the underlying graph’s factual accuracy, and Appendix H details why the current item pools are not yet large or adversarial enough to support strong claims either way. We do not identify a dual-use or malicious-use pathway specific to this release beyond those already inherent to publishing any biomedical knowledge graph or LLM evaluation harness.

6 Conclusion and Discussion

ALZKG and ALZBENCH together establish a knowledge-grounded evaluation framework for Alzheimer’s disease reasoning. ALZKG provides an open AD knowledge graph mined from 7,150 PMC full-text papers across five clinical layers, with every relation sentence-grounded and weighted by its true literature paper count, and ALZBENCH turns it into five plug-and-play reasoning tasks with a Graph-RAG retriever and a reproducible harness. Sentence-grounding and true paper-count weighting are necessary properties of a trustworthy graph, but our own audit (§2.5) shows they are not sufficient: an initial random-sample audit found only 13% of triplets fully correct, which we traced to concrete, fixable entity-disambiguation bugs; fixing them and re-auditing raised precision to 20% fully correct (60% correct-or-partial, up from 40%), a real improvement that still leaves further, differently-shaped failure modes (missing relation valence, category errors, unhandled negation) unaddressed. We present ALZKG as a reproducible, auditable, and now-once-iterated starting point rather than a validated ground-truth resource, with concrete remediation directions listed in Appendix H. Our intrinsic analysis shows that a compact 20–30-node, two-hop neighborhood is where Graph-RAG’s own coverage peaks, though absolute coverage stays low (at most 0.17) and a flat BM25 baseline over the same corpus reaches substantially higher coverage on the same metric—so we do not yet have evidence that the graph structure itself outperforms simple passage retrieval, only that it is a released, comparable option others can improve on.

ALZGRAPH raises questions we hope the community will study with the released harness: Can the extraction pipeline’s precision be raised substantially above the 20% our repeat audit measured—

through a systematic (rather than manually curated) medical-abbreviation resource, relation valence, negation handling, and AD-topic filtering (Appendix H)—without losing its sentence-grounded, true-paper-count honesty properties? Can Graph-RAG’s retrieval be improved to close, or explain, its large coverage gap against flat BM25 search on the same corpus? How much do anti-amyloid eligibility and ARIA-safety decisions improve when models retrieve genotype-dependent KG evidence rather than relying on parametric knowledge, once that evidence is itself more reliable? By releasing the mined graph with its true (if currently imprecise) statistics, two rounds of audited extraction samples, an intrinsic retrieval analysis benchmarked against a non-graph baseline, and a reproducible evaluation harness—rather than unverified numbers—we aim to provide a foundation for measuring, rather than assuming, progress on knowledge-augmented LLMs in real-world Alzheimer’s settings.

References

- [1] Marilyn S Albert, Steven T DeKosky, Dennis Dickson, et al. The diagnosis of mild cognitive impairment due to alzheimer’s disease. *Alzheimer’s & Dementia*, 7(3):270–279, 2011.
- [2] Randall J Bateman, Chengjie Xiong, Tammie LS Benzinger, et al. Clinical and biomarker changes in dominantly inherited alzheimer’s disease. *New England Journal of Medicine*, 367(9):795–804, 2012.
- [3] Céline Bellenguez, Fahri Küçükali, Iris E Jansen, et al. New insights into the genetic etiology of alzheimer’s disease and related dementias. *Nature Genetics*, 54(4):412–436, 2022.
- [4] Jacqueline S Birks and Richard J Harvey. Donepezil for dementia due to alzheimer’s disease. *Cochrane Database of Systematic Reviews*, (6), 2018.
- [5] Olivier Bodenreider. The unified medical language system (umls): integrating biomedical terminology. *Nucleic Acids Research*, 32(suppl_1):D267–D270, 2004.
- [6] Samantha Budd Haeberlein, Paul S Aisen, Frederik Barkhof, et al. Two randomized phase 3 studies of aducanumab in early alzheimer’s disease. *The Journal of Prevention of Alzheimer’s Disease*, 9(2):197–210, 2022.
- [7] Payal Chandak, Kexin Huang, and Marinka Zitnik. Building a knowledge graph to enable precision medicine. *Scientific Data*, 10(1):67, 2023.
- [8] Elizabeth H Corder, Ann M Saunders, Warren J Strittmatter, et al. Gene dose of apolipoprotein e type 4 allele and the risk of alzheimer’s disease in late onset families. *Science*, 261(5123):921–923, 1993.
- [9] Hejie Cui, Jiaying Lu, Ran Xu, et al. A review on knowledge graphs for healthcare: resources, applications, and promises. *Journal of Biomedical Informatics*, 2025.
- [10] Jeffrey Cummings, Liana Apostolova, Gil D Rabinovici, et al. Lecanemab: appropriate use recommendations. *The Journal of Prevention of Alzheimer’s Disease*, 10(3):362–377, 2023.
- [11] Darren Edge, Ha Trinh, Newman Cheng, et al. From local to global: a graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024.
- [12] Rita Guerreiro, Aleksandra Wojtas, Jose Bras, et al. Trem2 variants in alzheimer’s disease. *New England Journal of Medicine*, 368(2):117–127, 2013.
- [13] Harald Hampel, Sid E O’Bryant, José L Molinuevo, et al. Blood-based biomarkers for alzheimer disease: mapping the road to the clinic. *Nature Reviews Neurology*, 14(11):639–652, 2018.
- [14] Janna Hastings, Gareth Owen, Adriano Dekker, et al. Chebi in 2016: improved services and an expanding collection of metabolites. *Nucleic Acids Research*, 44(D1):D1214–D1219, 2016.
- [15] Clifford R Jack, David A Bennett, Kaj Blennow, et al. NIA-AA research framework: Toward a biological definition of alzheimer’s disease. *Alzheimer’s & Dementia*, 14(4):535–562, 2018.
- [16] Di Jin, Eileen Pan, Nassim Oufattole, et al. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021.
- [17] Sebastian Köhler, Michael Gargano, Nicolas Matentzoglou, et al. The human phenotype ontology in 2021. *Nucleic Acids Research*, 49(D1):D1207–D1217, 2021.

- 521 [18] Jean-Charles Lambert, Carla A Ibrahim-Verbaas, Denise Harold, et al. Meta-analysis of 74,046
522 individuals identifies 11 new susceptibility loci for alzheimer’s disease. *Nature Genetics*, 45
523 (12):1452–1458, 2013.
- 524 [19] Patrick Lewis, Ethan Perez, Aleksandra Piktus, et al. Retrieval-augmented generation for
525 knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, vol-
526 ume 33, pages 9459–9474, 2020.
- 527 [20] Gill Livingston, Jonathan Huntley, Andrew Sommerlad, et al. Dementia prevention, inter-
528 vention, and care: 2020 report of the lancet commission. *The Lancet*, 396(10248):413–446,
529 2020.
- 530 [21] Guy M McKhann, David S Knopman, Howard Chertkow, et al. The diagnosis of dementia due
531 to alzheimer’s disease. *Alzheimer’s & Dementia*, 7(3):263–269, 2011.
- 532 [22] Sebastian Palmqvist, Shorena Janelidze, Yakeel T Quiroz, et al. Discriminative accuracy of
533 plasma phospho-tau217 for alzheimer disease vs other neurodegenerative disorders. *JAMA*,
534 324(8):772–781, 2020.
- 535 [23] John R Sims, Jennifer A Zimmer, Cynthia D Evans, et al. Donanemab in early symptomatic
536 alzheimer disease: the trailblazer-alz 2 randomized clinical trial. *JAMA*, 330(6):512–527,
537 2023.
- 538 [24] Karan Singhal, Shekoofeh Azizi, Tao Tu, et al. Large language models encode clinical knowl-
539 edge. *Nature*, 620(7972):172–180, 2023.
- 540 [25] Reisa A Sperling, Paul S Aisen, Laurel A Beckett, et al. Toward defining the preclinical stages
541 of alzheimer’s disease. *Alzheimer’s & Dementia*, 7(3):280–292, 2011.
- 542 [26] Christopher H van Dyck, Chad J Swanson, Paul Aisen, et al. Lecanemab in early alzheimer’s
543 disease. *New England Journal of Medicine*, 388(1):9–21, 2023.

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A Related Work

Alzheimer’s evidence base. AD is now defined biologically through the ATN biomarker framework [15], with plasma and imaging biomarkers such as p-tau217 [22] and blood-based markers more broadly [13] entering routine use alongside established genetic risk factors (*APOE* ϵ 4 [8], *TREM2* [12], and autosomal-dominant genes tracked in longitudinal cohorts [2]) and a treatment landscape spanning long-standing cholinesterase inhibitors [4] and newer anti-amyloid antibodies [6, 23, 26]. This heterogeneity—genetics, fluid/imaging biomarkers, staging criteria, and a fast-moving treatment landscape with genotype-dependent safety constraints [10]—is exactly what motivates representing AD evidence as a typed, multi-hop graph rather than isolated facts.

Biomedical knowledge graphs. General-purpose biomedical KGs such as PrimeKG [7] and broader surveys of the space [9] provide wide disease coverage but are disease-agnostic by design: they do not encode the specific gene–biomarker–stage–treatment–outcome structure (e.g. genotype-dependent ARIA risk) that AD clinical reasoning requires. ALZKG takes the opposite trade-off, narrowing scope to one disease in exchange for a schema built around its actual clinical decision points.

Graph-RAG and knowledge-augmented LLMs. Retrieval-augmented generation grounds LLM outputs in external evidence [19], and graph-structured retrieval has been proposed to recover multi-hop reasoning chains that flat passage retrieval cannot [11]. LLMs have separately shown strong medical question-answering ability on general clinical knowledge [24]. ALZGRAPH’s Graph-RAG retriever combines these lines directly: a personalized-PageRank walk over ALZKG serializes literature-weighted reasoning paths as LLM context (§4.1). Unlike more elaborate retrievers, it has no semantic/embedding-based fallback (Appendix D.1)—a deliberate scope choice we discuss as a limitation, not a claimed advantage.

Clinical NLP benchmarks and LLM evaluation. General medical QA benchmarks such as MedQA-USMLE [16] test broad clinical knowledge but not the integrative, multi-hop, safety-critical reasoning AD care specifically demands; ALZBENCH reuses a dementia-filtered subset of MedQA-USMLE as one externally sourced task (T4, §3.2) precisely so that at least one task is not self-authored by the model it evaluates. More broadly, evaluating LLMs on clinical tasks surfaces evaluation-methodology pitfalls of its own—silent scoring bugs, self-authorship confounds, item-pool scale—that we found and disclose directly rather than only report headline numbers (Appendix H).

B AlzKG Construction Details

B.1 Ontology Sources and Coverage

Table 6 lists every source integrated into the recognition lexicon, the layer(s) it feeds, and how many entities it contributes. Unlike a single-tier vocabulary, ALZKG draws on two tiers per layer where applicable: a small, hand-curated *seed* (81 entities total, §2.3) naming well-established AD concepts, and, for the gene layer specifically, a much larger *recognition catalog* (the complete HGNC approved protein-coding gene set) that is not curated to AD relevance at all (§2.3); only entities that survive the sentence-grounded relation filter (§2.3, Table 1) enter the released graph.

Table 6: Ontology and guideline sources used in ALZKG construction. “Seed” entities are hand-curated (alzgraph/lexicon.py); “Catalog” entities are harvested programmatically (scripts/build_lexicon_from_ontologies.py) and only enter the released graph if they clear the sentence-grounded relation filter. Access reflects each source’s own public distribution terms as of this release; we report this descriptively rather than assert a specific license we have not independently verified.

Source	Layer(s)	Seed	Catalog	Access
NIA-AA ATN framework [15]	Biomarker, Stage	—	—	Published research framework
OMIM / ADSP / GWAS Catalog [3, 18]	Gene	25	—	Academic/open terms
HGNC	Gene	—	19,262	Public registry
MeSH	Biomarker, Stage	18 / 11	383 / 82	Public domain (NLM)
HPO [17]	Outcome	14	2,825	Open (publisher-stated)
ChEBI [14]	Treatment	—	5,702	Open (publisher-stated)
AAN / Alzheimer’s Association [10]	Treatment	13	—	Published guideline
UMLS [5]	cross-ontology linking	—	0 [†]	Free, registration req’d
Total		81	28,254	

[†] §2.5/Appendix B.5 discusses a lighter caseload-based substitute used in its place (merging entities that share a surface form) and a real bug this substitute introduced.

B.2 Entity Normalization Protocol

Recognition is a greedy, longest-n-gram token match (up to 8 tokens) over a compiled gazetteer, not a machine-learning NER model or an embedding-based fuzzy matcher: multi-word phrases take precedence over shorter substrings, gene symbols and other short, ambiguous strings are matched case-sensitively, and (post-fix, §2.5) any case-insensitive alias of 3 characters or fewer is dropped lexicon-wide. Surface forms are normalized to canonical entities through a curated synonym/abbreviation map (e.g. “Leqembi”→*Lecanemab*). Records from different ontology sources that share a surface form are merged into one canonical entity, but—since the extraction-precision audit found this merge rule itself could conflate unrelated concepts (§2.5)—only when the shared surface form is distinctive (multi-word or ≥ 5 characters) and the two records agree on layer. There is no semantic/embedding fallback for unresolved mentions: a genuine synonym or paraphrase absent from the curated alias map is simply not recognized, a recall gap distinct from (and not fixed by) the precision-focused audit and fix.

B.3 Relation Extraction Details

Every relation is rule-based: within a candidate sentence, a cross-layer entity pair induces a typed edge only if the sentence also contains one of that relation’s curated trigger phrases (Table 7 lists a representative subset of each relation’s full trigger set). ALZKG does not invoke an LLM-based

602 extraction pass in this release (§2.3); all 5,026 released triplets are rule-based, a deliberate scope
 603 choice to avoid reporting unverified edges rather than a claim that rule-based extraction is more
 604 accurate (§2.5/Appendix B.5 shows it is not, by itself, reliable either).

Table 7: Relation-extraction trigger phrases (representative subset of each relation’s full curated set; `scripts/build_kg_from_fulltext.py`). A relation fires only when a trigger substring co-occurs, in the same sentence, with a cross-layer entity pair of the listed layers.

Relation	Layer pair	Trigger phrases (examples)
risk_gene_for	gene → stage	risk, mutation, variant, carrier, predispos, suscep- tib
modulates_biomarker	gene → biomarker	modulat, regulat, increas, decreas, elevat, express
pharmacogenomic_consideration	gene → treatment	response to, genotype, contraindicat, tailor, metaboli
gene_associated_outcome	gene → outcome	associated with, predict, risk of, decline, progres- sion
characterized_by_biomarker	stage → biomarker	characterized by, positive, elevated, hallmark, bur- den
biomarker_guides_treatment	biomarker → treatment	eligib, candidat, indicat, screen, stratif, positive
predicts_outcome	biomarker → outcome	predict, correlat, prognos, marker of, indicative of
treated_with	stage → treatment	treat, therap, first-line, approved, prescrib
associated_outcome	stage → outcome	associated with, progress, decline, lead to, convert
has_outcome	treatment → outcome	resulted in, led to, adverse, efficac, side effect

605 B.4 Knowledge Graph Statistics

606 Table 8 complements the summary in Table 1. Two structural properties are worth stating explicitly
 607 because they differ from a typical mined biomedical KG: every released triplet is cross-layer by
 608 construction (co-occurring same-layer entity pairs never induce an edge, §2.3), so there is no within-
 609 layer/cross-layer split to report; and every triplet is rule-based (§B.3), so there is likewise no rule-
 610 based/LLM-based split.

Table 8: Detailed ALZKG statistics (post-fix graph, §2.5).

Statistic	Value	(%)
Total nodes	1,216	—
Total edges	5,026	100.0
Cross-layer edges	5,026	100.0
Rule-based edges	5,026	100.0
Median paper count	5	—
IQR paper count	3–10	—
Edges with paper count ≥ 10	1,304	25.9
Mean paper count	17.3	—

611 B.5 Extraction Precision Audit

612 §2.5 reports the headline numbers; this section gives the full protocol, both audits’ judgments, and
 613 every fix in detail.

614 We drew a reproducible random sample of 30 triplets (seed 42) from an earlier version of the re-
 615 leased graph (5,880 triplets, built from a then-current lexicon) and, for each, retrieved the actual
 616 corpus sentence used to emit it—the sentence whose recognized-entity list contains both the head
 617 and the tail—then judged whether that sentence supports the specific relation and direction claimed.

618 Of the 30: **4 (13%)** were fully correct, **8 (27%)** were partially or loosely supported (a real but
619 broader, weaker, or reversed-direction association than the relation claimed), and **18 (60%)** were
620 not supported by their own grounding sentence at all.

621 **Diagnosing and fixing the entity-disambiguation cases.** A recurring subset of the 18 unsupported
622 cases were *entity-disambiguation collisions*: the gene symbol *NOS1* matched the un-
623 related clinical abbreviation “NOS” (Not Otherwise Specified); the gene *MET* and the com-
624 pound L-lysine both matched one-letter amino-acid codes inside a protein-mutation notation
625 (“Lys670_Met671”); the gene *IGFALS* matched the disease abbreviation ALS as a bare sub-
626 string. Tracing these to their root cause in the entity-recognition code (alzgraph/lexicon.py,
627 scripts/build_lexicon_from_ontologies.py) surfaced two genuine software bugs, not just
628 curation gaps. (1) *Cross-concept alias merging*: the lexicon compiler merges two ontology records
629 whenever they share any casefolded surface form, intended to catch the same real-world concept
630 named twice across sources; but a short string like “MET”/“Met” collides by pure accident between
631 the *MET* gene symbol and an amino-acid alias of methionine, and the merge silently fused them
632 into one entity under the *gene* layer — so every case-insensitive mention of “met”/“methionine”
633 anywhere in the 361,201-sentence corpus was being recognized as the *MET* gene. We fixed this
634 by requiring the shared surface form to be distinctive (multi-word or ≥ 5 characters) and the two
635 records’ layers to match before merging. (2) *Order-dependent primary-symbol loss*: the raw HGNC
636 gene table lists many genes’ own official symbol as an *alias* of a different, unrelated gene (e.g.
637 *NANOS1* lists “NOS1” as one of its aliases); our original conflict-resolution logic let whichever
638 gene’s record appeared first in the file claim a contested string, so *NOS1* itself silently lost recogni-
639 tion of its own primary symbol to *NANOS1* purely due to file order. We fixed this by reserving every
640 gene’s own approved symbol for itself before resolving alias-level conflicts. On top of these two bug
641 fixes, we built a curated exclusion list (dictionary-word check via a standard word list, plus man-
642 ually verified clinical-abbreviation, institutional-acronym, and domain-generic-term collisions) that
643 drops a short gene alias — and, for a handful of audit-verified cases (*CBS*/corticobasal syndrome,
644 *NPS*/neuropsychiatric symptoms, *PGC*/Psychiatric Genomics Consortium), even a gene’s bare pri-
645 mary symbol — when it collides with a much more frequent non-gene meaning; *tau*/*TAU*, an official
646 MAPT alias that this literature uses constantly as ordinary vocabulary for the tau protein rather than
647 as a gene reference, is the largest single entry (removing it dropped *MAPT* out of the top-15 most-
648 mentioned entities entirely). All three fixes are entirely in code (no hand-edits to specific triplets),
649 so the resulting graph is rebuilt end-to-end from the same corpus and remains fully reproducible.

650 **Repeat audit on the fixed graph.** We rebuilt ALZKG with the fixed lexicon (Table 1: 5,026
651 triplets, down from 5,880) and drew a fresh, independent $n=30$ sample under the identical seed-42
652 protocol. Precision rises to **6 (20%)** fully correct, **12 (40%)** partial, and **12 (40%)** unsupported
653 — correct+partial rises from 40% to 60%, and the flatly-unsupported rate falls by a third (from
654 60% to 40%). This is a real, measured improvement from fixing concrete, named bugs, not a claim
655 of a solved problem: the repeat audit surfaces further failure modes distinct from entity disam-
656 biguation, which this fix does not address. *Lab-technique-as-biomarker category errors*: an overly
657 generic or mistyped MeSH lexicon entry (“Fluorescent Antibody Technique,” “Blotting, Western”) is
658 recognized as if it were a specific clinical biomarker rather than a laboratory method name.
659 *Glossary/abbreviations-list artifacts*: entities that only co-occur inside a paper’s front-matter ab-
660 breviations glossary (e.g. a list defining “TLR2” and “AD” as separate entries) are extracted as if the
661 paper’s body had asserted a relation between them. *Unhandled negation and hedging*: “no disease
662 specific treatments exist for frontotemporal dementia” was extracted as FTD TREATED_WITH, and
663 “its role in AD remains unclear” was extracted as a risk-gene claim — the trigger-phrase matcher
664 (§2.3) has no mechanism for an intervening negation or hedge. The original three failure modes also
665 persist in weaker form: *directionality loss* (a drug reported to *reduce* an outcome, e.g. cilostazol and
666 stroke, or intranasal insulin and memory impairment, is still stored identically to one that causes it)
667 and *off-topic leakage* (entities from unrelated case reports, e.g. an HIV cohort study or a bilingual-
668 language testing study, still enter the graph on lexicon co-occurrence alone). Full per-row judgments
669 for both audits are in data/alzkg/extraction_audit.json.

C AlzBench Dataset Details

C.1 Dataset Construction Protocols

T1 CDA. MCQ ($n=20$) and open-ended ($n=8$) items are hand-authored exam-style questions grounded in NIA-AA ATN staging, genetics, and treatment-safety knowledge (§3.2); MCQ distractors are clinically plausible but incorrect alternatives. **T2 CRG.** A local JSONL adapter mirrors the schema of a real memory-clinic report (patient history, cognitive assessment, biomarker panel $\rightarrow$ diagnostic impression); the public release ships one synthetic preview case rather than real ADNI/memory-clinic exports, which are access-restricted and not redistributed. **T3 BPM.** MCQs ($n=10$) are constructed from AAN/Alzheimer’s Association appropriate-use-criteria rules pairing an APOE genotype and ATN biomarker profile with a correct anti-amyloid-antibody eligibility decision, including ARIA-risk contraindications. **T4 TR.** Items ($n=16$) are filtered from real MedQA-USMLE [16] using a word-boundary dementia/AD-term match, followed by a manual relevance pass (Appendix H); gold answers are MedQA-USMLE’s own published answer key, not authored for this benchmark. Unlike some general-medicine QA benchmarks, no MMLU subset is included in this release. **T5 DRP.** Items ($n=20$) are randomly sampled real PMC abstracts from the AlzKG mining corpus (§2.3); there is no gold research plan to score against, so T5 is evaluated qualitatively on a subset (§4.5).

C.2 Gold Standard Annotation

No licensed clinician or domain-expert annotator was involved in constructing any AlzBench gold standard (Appendix H). T1/T3 gold answers were authored by the same LLM instance later evaluated on the Claude (direct) row of Table 5, a self-authorship confound discussed explicitly wherever that row is reported; T4’s gold answers come from MedQA-USMLE’s own external answer key rather than in-house authorship; T5 has no gold standard at all and is scored qualitatively. We report this plainly rather than describe an inter-annotator-agreement protocol we did not run.

C.3 Dataset Statistics

Table 9: ALZBENCH dataset statistics by task.

Task	Dataset	Instances	Gold standard	Split
T1 MCQ	Hand-authored	20	Hand-authored	Test only
T1 QA	Hand-authored	8	Hand-authored	Test only
T2 CRG	Synthetic preview	1	Synthetic	Test only (illustrative)
T3 BPM	AAN/AA-rule-derived	10	Rule-derived	Test only
T4 TR	MedQA-USMLE (dementia-filtered)	16	MedQA-USMLE answer key	Test only
T5 DRP	Real PMC abstracts	20 (2 used qualitatively)	None (qualitative)	Test only

D Graph-RAG Retriever Details

D.1 PPR Implementation

The retriever proceeds in three stages. First, the same dictionary-lexicon entity recognizer used in ALZKG construction (Appendix B.2) identifies seed entities in the query. Second, personalized PageRank is run from those seeds over the ALZKG adjacency (restart probability $\alpha=0.15$), using a pure-Python implementation with an optional `networkx`-backed fast path, producing a relevance score for every reachable node. Third, the highest-scoring nodes (up to the node budget) are kept and all distinct source-to-sink reasoning paths within the depth budget are enumerated and truncated to the highest-scoring paths, each serialized as `(head, relation[N papers], tail)` chains annotated with relation type and literature weight. Unlike a hybrid retriever, there is no semantic/embedding-based fallback: a query whose entities are not recognized by the lexicon retrieves nothing, a scope limitation we do not currently address.

D.2 Hyperparameter Settings

Table 10 lists the retriever’s hyperparameters and the values used to produce every result in this paper. Unlike a validation-tuned selection, the node-budget and depth defaults are *not* the settings that maximize gold-evidence coverage in §4.2’s own sweep (peak coverage is at 20 nodes / 2 hops, Table 3); we keep the wider 30-node / 4-hop default to preserve fuller reasoning-path context for the LLM rather than to optimize this one intrinsic metric, and report the resulting trade-off explicitly rather than silently select the metric-maximizing setting.

Table 10: Graph-RAG retriever hyperparameters. “Swept” columns report the range tested in §4.2’s ablation (Table 3); PPR restart probability and max serialized paths were not swept in this release.

Hyperparameter	Value used	Range swept (Table 3)
Max subgraph nodes	30	10, 20, 30, 40, 50
Max path depth	4	1, 2, 3, 4
PPR restart probability α	0.15	not swept
Max serialized paths	12	not swept

714 E Evaluation Metrics

715 All metrics below are pure-Python, standard-library implementations (`alzgraph/metrics.py`);
716 this release does not depend on an external NLP package (e.g. the official ROUGE/SacreBLEU pack-
717 ages or a BERTScore/roberta-large embedding model) or a working LLM-as-judge (Appendix H),
718 a deliberate no-external-dependency scope choice rather than an oversight.

719 **Top-1 Accuracy.** Exact match between the predicted option letter and the gold option letter, aver-
720 aged over items.

721 **ROUGE-L.** Longest-common-subsequence F1 between whitespace-tokenized prediction and refer-
722 ence text.

723 **Token-F1.** Bag-of-tokens overlap F1 (precision and recall over the multiset of tokens) between
724 prediction and reference.

725 **BLEU-1.** Clipped unigram precision between prediction and reference tokens (no brevity penalty).

726 **Drug Safety Score.** 1.0 if the selected option’s text does not contain any contraindicated-agent
727 substring for the given patient context (e.g. respecting ARIA exclusions), else 0.0.

728 **Guideline Concordance.** 1.0 if the selected option’s text contains any guideline-recommended
729 agent substring for the given context, else 0.0; defined in the released metrics code but not reported
730 as a headline number in this paper’s tables.

731 **KG Evidence Coverage (KGEC).** The fraction of the model’s answer tokens that also appear
732 among the tokens of the retrieved Graph-RAG paths—a lexical-overlap proxy for whether the an-
733 swer actively used the retrieved evidence, not a correctness measure.

734 **LLM-as-Judge.** Part of the task design (§3.2) for open-ended CDA/CRG alignment and DRP
735 scientific-validity/feasibility scoring, but no judge model, prompt, or agreement check exists in the
736 released metrics code; no number in this paper uses it (Appendix H).

F Prompts

Unlike a single shared system prompt across tasks, each AlzBench task uses its own role-specific system prompt (reproduced verbatim below, tasks/t*.py); a task’s user turn is the serialized Graph-RAG evidence (when the mode is graph_rag; a per-task literal prefix such as “AlzKG reasoning paths:” or “AlzKG context:”) prepended to the task’s own inputs (question/options, patient history, clinical scenario, or paper abstract, depending on the task).

T1 CDA — MCQ system prompt.

You are a cognitive neurologist taking an Alzheimer’s disease clinical decision exam. Select exactly one option letter (A, B, C, or D). Use NIA-AA biomarker (ATN) framing and guideline-consistent reasoning. Return only the option letter.

T1 CDA — open-ended QA system prompt.

You are a cognitive neurologist. Answer the clinical question in 2-4 concise sentences. Name relevant disease stages, ATN biomarkers (amyloid/tau/neurodegeneration), risk genes (e.g. APOE), treatments, contraindications, or outcomes when applicable.

T2 CRG system prompt.

You are a cognitive neurologist generating a diagnostic impression for a memory-clinic report. From the patient history, cognitive assessment, and biomarker panel, produce an impression that summarizes: (1) the cognitive syndrome and severity, (2) the ATN biomarker profile (amyloid, tau, neurodegeneration) and whether it supports Alzheimer’s disease, (3) the most likely diagnosis and stage, and (4) management or follow-up recommendations. Be concise, evidence-grounded, and clinically safe.

T3 BPM system prompt.

You are a clinical neurologist specializing in Alzheimer’s disease pharmacotherapy. Select the single most appropriate treatment from A-D using APOE genotype, ATN biomarker status, and AAN / Alzheimer’s Association appropriate-use-criteria safety reasoning (including ARIA risk). Return only the option letter.

T4 TR system prompt.

You are a clinical neurologist managing Alzheimer’s disease and related dementias. Select the safest guideline-consistent treatment option from A-D, considering contraindications, ARIA risk for anti-amyloid antibodies, comorbidities, and disease stage. Return only the option letter.

T5 DRP system prompt.

You are an Alzheimer’s disease clinical researcher. Given a paper abstract, generate: 1. a focused, novel research question, 2. a study design rationale (cohort, biomarkers, endpoints), 3. the required evidence or data and key feasibility considerations. The plan must be feasible, clinically meaningful, and grounded in known gene-biomarker-stage-treatment-outcome evidence (ATN framework).

G Additional Experimental Results

Table 11 consolidates every pre-fix vs. post-fix number in this paper (§2.5/Appendix B.5) in one place, for a reader who wants the full before/after picture without cross-referencing each subsection individually.

Table 11: Every measured pre-fix vs. post-fix number in this paper, in one place (§2.5, §4.3, §4.4). “—” marks a quantity not remeasured post-fix (Claude (direct), Appendix H) or unaffected by construction (BM25/No-RAG, which do not consult ALZKG’s entity/edge content the way Graph-RAG and the KG-only baseline do).

Quantity	Task/setting	Pre-fix	Post-fix
Entities	—	1,301	1,216
Triples	—	5,880	5,026
Audit: fully correct	$n=30$	4 (13%)	6 (20%)
Audit: partial	$n=30$	8 (27%)	12 (40%)
Audit: unsupported	$n=30$	18 (60%)	12 (40%)
KG-only baseline accuracy	T1 CDA	0.55	0.50
KG-only baseline accuracy	T3 BPM	0.30	0.30
KG-only baseline accuracy	T4 TR	0.25 (chance)	0.375
KG-only accuracy (grounded)	T4 TR	0.23 (below chance)	0.42
Graph-RAG accuracy	T3, Qwen3-8B	0.20	0.30
Graph-RAG accuracy	T4, Qwen3-8B	0.63	0.44
Graph-RAG accuracy	T4, Llama-3.2-3B	0.3125	0.19
Retrieval ablation gold coverage (peak)	node/depth sweep	0.10–0.17	0.10–0.17 (unchanged)
BM25 gold coverage	non-graph baseline	— (not run pre-fix)	0.567

H Limitations and Future Work

Item scale and self-authorship. T1’s MCQ/QA and T3’s items are curated, hand-authored exam-style questions (20, 8, and 10 items respectively after this revision’s expansion from 10/3/5); T4 draws 16 items from real MedQA-USMLE via a dementia-term filter; T5 currently offers 20 real abstracts but only 2 have been used for the qualitative case study; T2 has exactly one item (§4.5) and is, in effect, unevaluated pending access to a non-restricted clinical-report dataset. These are still small samples for a benchmark claim. The self-authorship concern is specific to the *Claude (direct)* row: T1/T3 were authored by the same model instance that later answered them, which we believe explains its ceiling accuracy there more than any special capability of Graph-RAG. It does not apply to the three local open-source models in Table 5 (none of which authored any item), and their genuinely mixed, sub-ceiling results are correspondingly more informative, though still measured on the same small item pools. A more discriminating benchmark would (i) grow each item pool by an order of magnitude, (ii) source items independently of the evaluated model (as T4 already does), and (iii) deliberately include harder, rarer, or adversarially constructed cases (atypical presentations, conflicting guideline edge cases, recently updated appropriate-use criteria) where a model’s parametric knowledge is less likely to already be saturated.

The Claude (direct) row is not rerun post-fix. Table 5’s Claude (direct) row is reported as originally measured, on the pre-fix graph (§2.5), for both the No-RAG and Graph-RAG columns; it was not re-answered on the post-fix graph the way the three open-weight rows were (§4.4). Every one of its conditions already sits at ceiling accuracy, attributed above to self-authorship (T1/T3) or same-model-family items (T4) rather than retrieval quality, so we judged a full manual re-answer unlikely to change the reported numbers and prioritized rerunning the open-weight rows instead, which are not ceiling-bound and are the table’s more informative comparison. We flag this rather than silently present a mixed-vintage table as if every cell were measured under identical conditions.

Option-letter extraction from free-form text. Our answer-extraction routine originally case-folded a model’s completion and returned the *first* standalone A/B/C/D match, which is fine for a short direct answer but silently wrong for a long chain-of-thought completion: the lowercase article “a” folds to “A” and can be matched well before a reasoning model states its actual conclusion. This first surfaced as an implausible Qwen3-8B T3 Graph-RAG accuracy built mostly out of such spurious matches; spot-checking the raw reasoning traces confirmed it (an early, uncorrected version of Table 5 reported T3 Graph-RAG 0.50 for Qwen3-8B). We fixed the extractor to prefer the *last* explicit “answer/option/choice is/might be X” phrase — robust to a model second-guessing itself mid-trace — falling back to the last case-sensitive bare letter only otherwise; Table 5’s Qwen3-8B numbers use the corrected extractor throughout (T3 Graph-RAG 0.30 on the current, post-fix graph, §2.5; 0.20 on the pre-fix graph reported in an earlier version of this paper). We flag this as a limitation because it is exactly the class of silent scoring bug that is easy to miss without inspecting raw model output, and a benchmark harness intended for others to trust should say so plainly rather than only report the corrected number.

Term-filtered external items are noisy. T4’s dementia-term filter, even after switching to word-boundary matching (an early version matched “mci” inside “triamcinolone” and bare “amyloid” inside unrelated systemic-amyloidosis questions), still required a manual relevance pass that dropped 5 of 21 automatically filtered items because the matched term appeared only in a distractor option or a patient’s incidental history rather than the tested clinical reasoning; scaling T4 will need a more precise filter or classifier rather than manual review.

Retriever determinism. An earlier version of the retriever iterated the graph’s node set directly in several places (seed matching, PageRank state, and top- k neighborhood selection); because Python’s set iteration order depends on a per-process string-hash seed, two runs of the identical query against the identical graph could silently keep a different tied path or seed order, occasionally changing the serialized Graph-RAG evidence (and, in one case we caught during evaluation, changing a downstream answer). We fixed this by iterating over a sorted node order everywhere ranking or state construction depends on it, and verified byte-identical output across five repeated runs with randomized hash seeds for both the retriever and the KG-only baseline; released code reflects this fix.

Extraction precision (see also §2.5/Appendix B.5). Our first manual audit found only 13% of a random triplet sample fully correct and 60% unsupported by their own grounding sentence. We traced most of this to two genuine software bugs (a cross-concept alias-merge that fused unrelated ontology entries sharing a short casefolded string, and an order-dependent conflict-resolution pass that could strip a gene’s own primary symbol) plus a curatable set of short-symbol collisions, fixed all three, rebuilt the graph, and re-audited: precision rose to 20% fully correct / 40% unsupported, a real but partial improvement. This release still ships with three further, structurally different failure modes the fix does not address, all found in the repeat audit: no valence on the relation schema (so “improves” and “causes” are stored identically); lab-technique-as-biomarker category errors from overly generic or mistyped lexicon entries; and glossary/abbreviations-list co-occurrence and unhandled sentence-level negation both being extracted as if they were asserted relations. Concrete remediation directions, none yet implemented in this release: extend the exclusion-list approach of §2.5 with a proper medical-abbreviation resource rather than a manually curated list, since residual collisions of that exact type should be expected (the repeat audit itself surfaced three more instances – IGF, PGC, TAU – that the first fix’s list had not anticipated); add a signed valence field to each relation rather than inferring it from the trigger phrase alone; add a lightweight negation/hedge detector (e.g. reject a sentence containing “no”, “not”, “unclear”, or “remains unknown” within the trigger-phrase window) and a check that a matched entity pair is not confined to a front-matter abbreviations list; require the source sentence (or its paper) to also match an AD-anchored topic filter, not just contain a lexicon entity, to catch off-topic leakage that the audit still finds; and add allele-level entities (e.g., *APOE* $\varepsilon 2/\varepsilon 3/\varepsilon 4$) so genuinely protective and risk variants of the same gene are not collapsed onto one node, which is a real gap given the paper’s own headline use case (genotype-dependent ARIA safety, §1) depends on exactly this distinction.

The non-graph retrieval comparison is intrinsic only. §4.2’s BM25 comparison measures gold-token coverage, not downstream task accuracy: we did not re-run the full No-RAG/Graph-RAG language-model comparison (§4.4) under a third condition where models are given BM25-retrieved passages instead of Graph-RAG paths. Whether flat retrieval’s higher coverage translates into better model-facing accuracy, worse accuracy (e.g., from noisier or less structured context), or no difference remains an open, measurable question with the harness as released.

No clinician review. No licensed clinician reviewed the hand-authored T1/T2/T3 items, the mined ALZKG triplets audited in §2.5, or any model output for clinical accuracy or safety. Every correctness judgment in this paper—item design, the extraction audit, drug-safety scoring—was made by an LLM or by the paper’s authors, not by a domain expert blinded to the system under test. This matters more than usual here because the benchmark is explicitly framed around safety-critical decisions (ARIA eligibility, contraindications).

LLM-as-judge is unimplemented. §3.2/§3.3 describe an LLM-as-judge dimension for open-ended CDA, CRG, and DRP scientific-validity/feasibility scoring, but no judge model, prompt, or agreement check exists in the released metrics code, and no number in this paper uses it. All open-ended results reported here use only lexical overlap metrics (ROUGE-L, BLEU-1, Token-F1), which are known to correlate imperfectly with clinical correctness.

No societal-harm-specific safeguards beyond standard release practices. ALZKG and ALZBENCH are a research artifact for evaluating retrieval-augmented reasoning, not a clinical decision-support product; none of the released code or data should be used to make real treatment decisions without a licensed clinician, and the KG’s coarse, sentence-level relation extraction can and does surface spurious associations (§2.3) that would need clinical review before any downstream use.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and §1 state exactly what is measured (mined KG statistics, two

rounds of manual extraction-precision audit with the bugs found and fixed between them, an intrinsic retrieval analysis benchmarked against a non-graph baseline, a deterministic KG-only baseline, and a blind No-RAG vs. Graph-RAG comparison run against four models: Claude direct and three local open-weight models), including the unflattering findings (13%→20% audited-triplet precision, still well short of reliable; BM25 exceeding Graph-RAG’s own coverage metric), rather than implying either a more reliable graph or a broader model comparison than we ran.

2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [Yes]

Justification: Appendix H discusses item-pool scale (including T2’s unevaluated, $n=1$ status), self-authorship of the T1/T3 items, T4’s term-filter noise, a retriever-determinism bug we found and fixed, the extraction-precision audit’s findings and remediation directions, the intrinsic-only scope of the non-graph retrieval comparison, the absence of any clinician review, and the unimplemented LLM-as-judge dimension.

3. Theory Assumptions and Proofs

Question: Does the paper include theoretical results requiring stated assumptions and proofs?

Answer: [NA]

Justification: The paper is empirical/systems-oriented (a knowledge graph, a benchmark, and a retriever); it contains no theorems.

4. Experimental Result Reproducibility

Question: Does the paper disclose all information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: The KG-only baseline is deterministic and reproducible with a single command per dataset (§4.3); the Claude-direct row’s exact prompts, queue, and cache format are described in §4.4 and implemented in the released evaluation code; the extraction-precision audit’s triplet sample is reproducible with a fixed random seed and a released script, and the non-graph retrieval baseline is a released, deterministic script over the same corpus and probes as the Graph-RAG ablation; the scripts used to construct each task’s dataset are released alongside the data they produced.

5. Open Access to Data and Code

Question: Does the paper provide open access to the data and code, with sufficient instructions?

Answer: [Yes]

Justification: Code and data are included as supplementary material for review and will be published at a public repository on acceptance (identity withheld for anonymity per Appendix H); a manifest included in the release maps every paper component to its release code.

6. Experimental Setting/Details

Question: Does the paper specify all training/evaluation details needed to understand the results?

Answer: [Yes]

Justification: §4.1 (Setup) gives PPR hyperparameters, neighborhood/path budgets, and prompting protocol; §4.3–4.4 give the exact evaluation commands and item counts per task.

7. Experiment Statistical Significance

Question: Does the paper report error bars or other measures of statistical significance?

Answer: [No]

Justification: Item counts (8–20 per task) are too small for meaningful confidence intervals on a single-model, single-run comparison; we report raw accuracy/counts throughout and flag small- n explicitly rather than presenting unsupported error bars (Appendix H).

8. Experiments Compute Resources

Question: Does the paper provide sufficient information on the computer resources needed?

Answer: [Yes]

Justification: KG mining, PPR retrieval, the KG-only baseline, and the Claude-direct evaluation run on CPU in well under an hour on a single machine, with no distributed training involved (§4.2 reports mean per-query retrieval latency, 30–70 ms). The three local-model rows in Table 5 (§4.4) additionally used one GPU (a single NVIDIA H200 or RTX 6000, model-dependent, of a shared multi-GPU machine) via a local Ollama server, with each model’s full 8-condition run (4 tasks $\times$ 2 modes) completing in a few minutes to under an hour depending on model size and whether the model’s chain-of-thought mode was enabled.

9. **Code Of Ethics**
 Question: Does the research conform to the NeurIPS Code of Ethics?
 Answer: [Yes]

10. **Broader Impacts**
 Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?
 Answer: [Yes]
 Justification: See §*Broader Impact* above.

11. **Safeguards**
 Question: Does the paper describe safeguards for responsible release of models/data with a high risk of misuse?
 Answer: [NA]
 Justification: The release is a knowledge graph mined from public open-access literature and a benchmark harness, not a generative model or a dataset of sensitive personal data (the private ADNI/memory-clinic-derived T2 adapter is explicitly not redistributed, §3.2); we do not see a specific high-risk-of-misuse vector beyond those already discussed in §*Broader Impact*.

12. **Licenses**
 Question: For assets used, are the license and terms of use respected and properly cited?
 Answer: [Yes]
 Justification: Ontology/guideline resources (NIA-AA, OMIM, GWAS Catalog, MeSH, HPO, ChEBI, UMLS, AAN/AA) and the MedQA-USMLE dataset [16] are cited at first use (§2, §2.2); the release’s own license is Apache-2.0.

13. **Assets**
 Question: Are new assets introduced in the paper well documented?
 Answer: [Yes]
 Justification: ALZKG, ALZBENCH, and the Graph-RAG retriever are documented in §2–3 and in the accompanying documentation released with the code, including provenance/honesty notes on how each artifact was produced.

14. **Crowdsourcing and Research with Human Subjects**
 Question: Does the paper involve crowdsourcing or research with human subjects?
 Answer: [NA]
 Justification: No human-subjects data collection was performed; all clinical text is either literature (PMC open-access articles) or de-identified, publicly released benchmark items (MedQA-USMLE) or curated exam-style items.

15. **IRB Approvals**
 Question: Does the paper describe potential risks to participants and IRB approval, if applicable?
 Answer: [NA]
 Justification: No human subjects were involved (see previous item); T2’s private ADNI/memory-clinic adapter operates only on data the institution already holds under its own IRB-approved protocols and is not redistributed by this release.

16. **Declaration of LLM Usage**
 Question: Does the paper describe the usage of LLMs in the core methodology of the research?
 Answer: [Yes]
 Justification: LLMs are a core, disclosed part of the methodology in §4.4: Claude (Sonnet 5) both authored some T1/T3 benchmark items and, separately, served as the evaluated

1001 model for the Claude-direct row, blind to gold labels during the latter role; three open-
1002 weight local models (Llama-3.2-3B, Gemma3-4B, Qwen3-8B) additionally served as eval-
1003 uated models proper, with no authorship role. These roles and their implications for the
1004 reported ceiling accuracy (Claude) versus genuinely mixed accuracy (local models) are
1005 discussed explicitly in §4.4 and Appendix H rather than left implicit.